2.5 Viète's Product

WORKSHEET

The question

STEP III 2022, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What is the double-angle formula for sinh? (If you are not sure, write $\sinh 2y$ and $\sinh y \cosh y$ in terms of exponentials and compare.)

STAGE 1 Halving repeatedly

Covers: Part (i)

Ask yourself The case $n = 1$ is the double-angle formula $\sinh 2y = 2\sinh y \cosh y$, with $y = \frac{x}{2}$.

Pause and try If the formula holds for n, its last factor is $\sinh(x/2^n)$. Apply the double-angle formula to that factor. For the deduction, divide by x and rearrange.

STAGE 2 The limit

Covers: Part (ii)

Ask yourself From the series for $\sinh y$, what does $\frac{\sinh y}{y}$ tend to as $y \to 0$?

Pause and try Let $n \to \infty$ in part (i) with x fixed. What happens to $\frac{x/2^n}{\sinh(x/2^n)}$?

STAGE 3 A formula for $\ln 2$

Covers: Part (iii)

Ask yourself With $x = \ln 2$, $e^{x/2} = \sqrt{2}$. Work out $\cosh \frac{x}{2}$ and $\cosh \frac{x}{4}$, and simplify.

Pause and try Find the general pattern for $\cosh(x/2^k)$. When you multiply the factors together, what do the powers of 2 combine to? Also work out $\frac{\sinh x}{x}$.

STAGE 4 Viète's formula

Covers: Part (iv)

Ask yourself Part (iv) has π in it, so it must come from ordinary trigonometry. Do parts (i) and (ii) still work with sin and cos in place of sinh and cos?

Pause and try Choose x to make $\frac{\sin x}{x} = \frac{2}{\pi}$, and work out the first few factors. The half-angle formula $\cos\frac{\theta}{2} = \sqrt{\frac{1+\cos\theta}{2}}$ will help.

What's really going on

Archimedes' polygons

Viète's formula has a geometric meaning that goes back to Archimedes. A regular polygon with 2^n sides inscribed in a circle of radius 1 has area $A_n = 2^{n-1}\sin\frac{2\pi}{2^n}$. Doubling the number of sides multiplies the area by

$$\frac{A_{n+1}}{A_n} = \frac{2^n \sin(\pi/2^n)}{2^{n-1}\cdot 2\sin(\pi/2^n)\cos(\pi/2^n)} = \frac{1}{\cos(\pi/2^n)}.$$

Starting from the square ($A_2 = 2$) and doubling for ever, the area tends to the area of the circle, π. So $\pi = 2 \div (\cos\frac{\pi}{4}\cos\frac{\pi}{8}\cos\frac{\pi}{16}\cdots)$, which is Viète's formula. Each factor in the product is the ratio of the areas of two successive polygons in Figure 2.5.1.

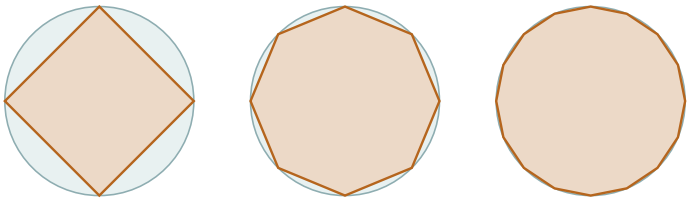


Figure 2.5.1 Regular polygons with 4, 8 and 16 sides inscribed in a circle of radius 1. Each doubling of the number of sides divides the area by one more factor of Viète's product, and the areas increase towards π.

Infinite products

Viète's product was the first infinite product in mathematics. Others followed: John Wallis found $\frac{\pi}{2} = \frac{2}{1}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdot\frac{4}{5}\cdot\frac{6}{5}\cdot\frac{6}{7}\cdots$ in 1655, and Euler's product for $\frac{\sin x}{x}$ appeared in Section 2.4. At university you will learn when an infinite product converges. Roughly, $\prod(1+a_n)$ with small positive a_n behaves like $\sum a_n$, because $\ln(1+a_n) \approx a_n$. Here $1-\cos(\pi/2^n) \approx \frac{1}{2}(\pi/2^n)^2$, which shrinks by a factor of 4 each time, so the product converges quickly.

How quickly?

After n factors the partial product equals $\dfrac{1}{2^n \sin(\pi/2^{n+1})}$ exactly, and its error is about $\frac{\pi^2}{6\cdot 4^{n+1}}$ times $\frac{2}{\pi}$. Each new factor makes the answer about 4 times more accurate, gaining about 0.6 of a decimal digit. Ten factors give $2/\pi$ correct to six decimal places. That is good, but nothing like as fast as the methods of Section 2.6, which double the number of correct digits at each step.

Try this next

1. **The 1990 version.** Try STEP I 1990, Question 8 (step.maths.org/questions/90-s1-q8). It reaches Viète's product by a different route from this section, using nothing beyond the double-angle formula for sin. Which approach do you find more natural?

2. **Another angle.** Use $x = \frac{\pi}{3}$ in the product for $\frac{\sin x}{x}$ to write $\frac{3\sqrt{3}}{2\pi}$ as an infinite product, and simplify the first two factors.

3. **How many factors?** Use a calculator to find how many factors of Viète's product are needed to give π correct to 4 decimal places. Explain why each extra factor gains about 0.6 of a digit.

4. **A formula for $\ln 3$.** Repeat part (iii) with $x = \ln 3$ to write $\frac{1}{\ln 3}$ as an infinite product. How many factors are needed to give $\frac{1}{\ln 3}$ to two decimal places? Why does this product converge more slowly than Viète's?

Further reading

- **Petr Beckmann, *A History of* π** (St Martin's Press). Archimedes, Viète, Wallis and the rest, told with gusto.
- **Jonathan Borwein & Peter Borwein, *Pi and the AGM*** (Wiley). The serious mathematics of computing π, including the method of Section 2.6. Chapter 1 is accessible with this chapter behind you.
- **Eli Maor, *Trigonometric Delights*** (Princeton University Press). Contains a chapter on Viète and his product.

The full guided solution is in Section 2.5:
https://beyondstep.keithmcnulty.org/ch02/viete.html